

MSDM5004

Numerical Methods and Modeling in Science
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Lecture 13

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3. Elliptic PDEs

A model problem

$$u_{xx} + u_{yy} + f(x, y) = 0, \quad (x, y) \in \Omega, \quad (1)$$

$$u = 0, \quad (x, y) \in \partial\Omega, \quad (2)$$

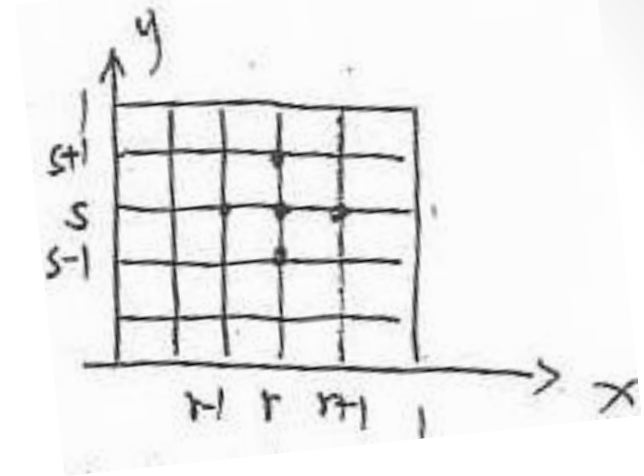
$$\Omega := (0, 1) \times (0, 1)$$

Notations: $u_{xx} = \frac{\partial^2 u}{\partial x^2}, \quad u_{yy} = \frac{\partial^2 u}{\partial y^2}$

An Important Property: **Maximum Principle**

If $f(x, y) \leq 0$, then the maximum of Eq.(1) is attained on the domain boundary.

Uniform mesh $\Delta x = \Delta y = 1/J$



Five-point scheme

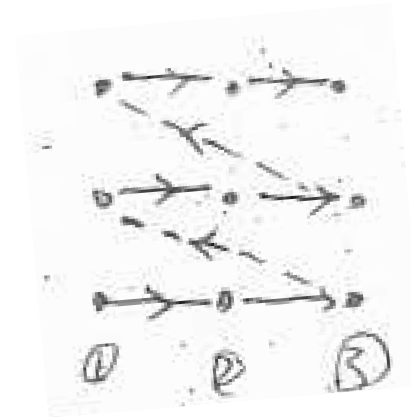
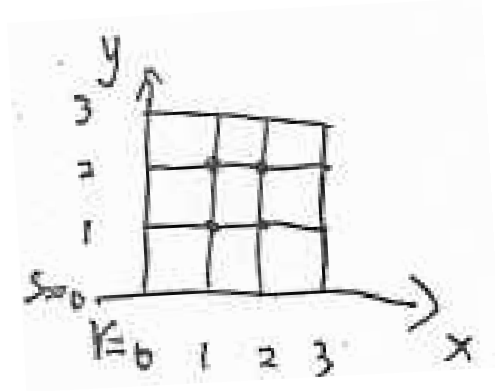
$$\frac{U_{r+1,s} + U_{r-1,s} + U_{r,s+1} + U_{r,s-1} - 4U_{r,s}}{(\Delta x)^2} + f_{r,s} = 0.$$

$$r = 1, 2, \dots, J-1 \quad s = 1, 2, \dots, J-1$$

Obtained by central difference approximations of u_{xx} and u_{yy}

$$\frac{U_{r+1,s} - 2U_{r,s} + U_{r-1,s}}{(\Delta x)^2} + \frac{U_{r,s+1} - 2U_{r,s} + U_{r,s-1}}{(\Delta y)^2} + f_{r,s} = 0$$

e.g. $J=3$



Natural order:

$$U_{1,1}, U_{2,1}, U_{3,1}, \dots, U_{J-1,1}, U_{1,2}, U_{2,2}, \dots, U_{J-1,2}, \dots, U_{1,J-1}, U_{2,J-1}, \dots, U_{J-1,J-1}$$

$$\begin{bmatrix} -4 & 1 & 1 & 0 \\ 1 & -4 & 0 & 1 \\ 1 & 0 & -4 & 1 \\ 0 & 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} U_{1,1} \\ U_{2,1} \\ U_{1,2} \\ U_{2,2} \end{bmatrix} = \begin{bmatrix} -f_{1,1}(\Delta x)^2 - g_{1,0} - g_{0,1} \\ -f_{2,1}(\Delta x)^2 - g_{2,0} - g_{3,1} \\ -f_{1,2}(\Delta x)^2 - g_{0,2} - g_{1,3} \\ -f_{2,2}(\Delta x)^2 - g_{3,2} - g_{2,3} \end{bmatrix}$$

Truncation error

$$T_{r,s} = \frac{u(x_{r+1}, y_s) + u(x_{r-1}, y_s) + u(x_r, y_{s+1}) + u(x_r, y_{s-1}) - 4u(x_r, y_s)}{(\Delta x)^2} + f(x_r, y_s)$$

$$= \frac{1}{12}(\Delta x)^2 (u_{xxxx} + u_{yyyy})_{r,s} + O((\Delta x)^4)$$

$$|T_{r,s}| \leq T := \frac{1}{12}(\Delta x)^2 (M_{xxxx} + M_{yyyy})$$

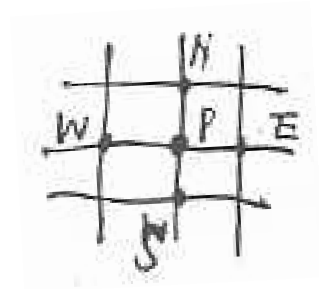
$$M_{xxxx} = \max_{\Omega} |u_{xxxx}|, \quad M_{yyyy} = \max_{\Omega} |u_{yyyy}|$$

Remark: Convergence of this scheme can be proved by using maximum principle of the numerical scheme.

Iterative methods for solving the linear system

Write the five-point scheme as

$$\frac{U_E - 2U_P + U_W}{(\Delta x)^2} + \frac{U_N - 2U_P + U_S}{(\Delta x)^2} + f_P = 0.$$



$$4U_P - (U_E + U_W + U_N + U_S) = (\Delta x)^2 f_P.$$

Moving the values on the boundary to the right

$$\tilde{c}_P U_P - [\tilde{c}_E U_E + \tilde{c}_W U_W + \tilde{c}_N U_N + \tilde{c}_S U_S] = b_P$$

e.g., if E is on the boundary, $U_E = g_E$,

$$4U_P - (0 + U_W + U_N + U_S) = (\Delta x)^2 f_P + g_E.$$

Thus $\tilde{C}_P = 4$, $\tilde{C}_E = 0$, $\tilde{C}_W = \tilde{C}_N = \tilde{C}_S = 1$, $b_P = (\Delta x)^2 f_P + g_E$.

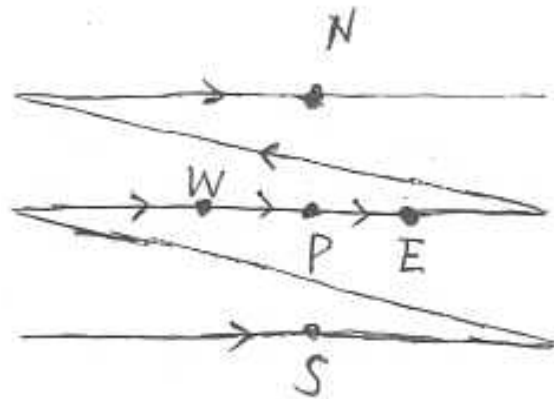
Jacobi iteration

$$U_P^{(n+1)} = (1/\tilde{c}_P)[b_P + \tilde{c}_E U_E^{(n)} + \tilde{c}_W U_W^{(n)} + \tilde{c}_N U_N^{(n)} + \tilde{c}_S U_S^{(n)}]$$

Gauss-Seidel iteration

$$U_P^{(n+1)} = (1/\tilde{c}_P)[b_P + \tilde{c}_E U_E^{(n)} + \tilde{c}_W U_W^{(n+1)} + \tilde{c}_N U_N^{(n)} + \tilde{c}_S U_S^{(n+1)}]$$

Natural order:



SOR

$$U_P^{(n+1)} = (1 - \omega)U_P^{(n)} + (\omega/\tilde{c}_P)[b_P + \tilde{c}_E U_E^{(n)} + \tilde{c}_W U_W^{(n+1)} + \tilde{c}_N U_N^{(n)} + \tilde{c}_S U_S^{(n+1)}]$$

Boundary conditions

Dirichlet boundary condition :

$$u(x, y) = g(x, y), \quad (x, y) \in \partial\Omega.$$

Neumann boundary condition :

$$\frac{\partial u}{\partial n} = g, \quad (x, y) \in \partial\Omega.$$

Robin boundary condition :
(mixed boundary condition)

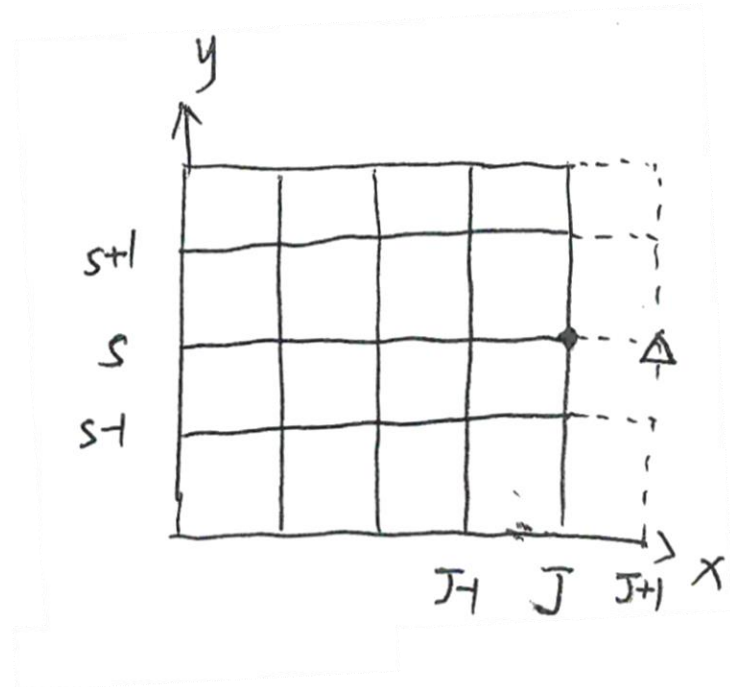
$$\alpha_1 \frac{\partial u}{\partial n} + \alpha_0 u = g, \quad (x, y) \in \partial\Omega.$$

$$\alpha_0, \alpha_1 \geq 0, \alpha_0 + \alpha_1 > 0.$$

$$u_{xx} + u_{yy} + f(x, y) = 0, \quad (x, y) \in \partial\Omega,$$

$$\alpha_1 \frac{\partial u}{\partial n} + \alpha_0 u = g, \quad (x, y) \in \partial\Omega,$$

where $\Omega = (0, 1) \times (0, 1)$.



Consider the boundary condition at the point (x_J, y_s) .

Add one more point (x_{J+1}, y_s) with solution value $U_{J+1,s}$ on it.

The boundary condition can be approximated by

$$\alpha_1 \frac{U_{J+1,s} - U_{J-1,s}}{2\delta x} + \alpha_0 U_{J,s} = g_{J,s}. \quad (\text{second order})$$

Assuming that the equation also holds on (x_J, y_s) , we have

$$\frac{U_{J+1,s} - 2U_{J,s} + U_{J-1,s}}{(\Delta x)^2} + \frac{U_{J,s+1} - 2U_{J,s} + U_{J,s-1}}{(\Delta y)^2} + f_{J,s} = 0.$$

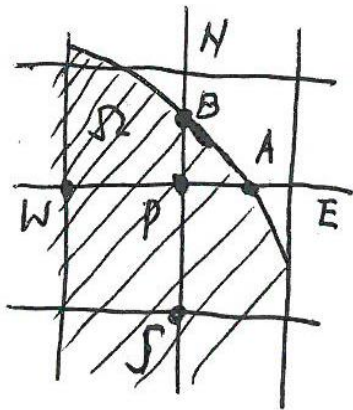
We can use these two equations to eliminate $J_{J+1,s}$.

General boundaries

$$u_{xx} + u_{yy} + f = 0, \quad (x, y) \in \Omega,$$

$$u = g, \quad (x, y) \in \partial\Omega,$$

where Ω has curved boundary $\partial\Omega$.



Consider the equation at point P

$$PA = \theta \Delta x, \quad 0 < \theta < 1$$

$$U_A = \left[u + \theta \Delta x u_x + \frac{1}{2} (\theta \Delta x)^2 u_{xx} + \cdots \right]_P$$

$$U_W = \left[u - \Delta x u_x + \frac{1}{2} (\Delta x)^2 u_{xx} + \cdots \right]_P$$

Eliminating u_x , we have

$$[U_{xx}]_P \approx \frac{U_A + \theta u_W - (1 + \theta) U_P}{\frac{1}{2} \theta (1 + \theta) (\Delta x)^2} \quad (\text{first order})$$

Finite Element Method

- Another class of numerical methods for solving elliptic equations are the **finite element methods**
- Based on variational formulation of the equations

